

Capture the Flag 3: Data-Driven UI

Goal: Separate application data from the UI and handle multiple pieces of data in a usable interface.

Instructor Note

These activities require your own problem-solving and research skills.

DO NOT use AI to generate the code for you.

You may use AI to:

- Research what a Dart/Flutter feature does
- Understand syntax
- **Explain** an error (not **fix** your error)
- Clarify documentation

You may **NOT** ask AI to generate the solution/code for a flag.

🚩 Flag 0 — Create Your Feature Branch

Objective: Create a feature branch before modifying your application.

Create:

`feature/data-ui`

Screenshot:

- Terminal showing your current branch

```
braiz@Mooney MINGW64 ~/MynewFlutterapp/IT314AB_ORAIZ (feature/widgets)
$ git status
On branch feature/widgets
nothing to commit, working tree clean

braiz@Mooney MINGW64 ~/MynewFlutterapp/IT314AB_ORAIZ (feature/widgets)
$ git checkout -b feature/data-ui
Switched to a new branch 'feature/data-ui'

braiz@Mooney MINGW64 ~/MynewFlutterapp/IT314AB_ORAIZ (feature/data-ui)
$ |
```

Flag 1 — Identify the Data

Objective: Identify the appropriate Dart data type for the information in your profile.

Research basic Dart data types and identify at least **four different types** that could be used in your application.

Examples:

String
int
double
bool

Create a table identifying the information and its appropriate data type.

Create:

- Your identified data
- The corresponding data types

IDENTIFIED DATA	DATA TYPE
Height	Double
Date	DateTime
Course And section	String
Age	Int
Student status	Bool
Name	String
Hobby	String
Favorite Foods	String
Favorite Movie	String
Favorite Game	String



Flag 2 — Create Your Data

Objective: Create appropriately typed variables for your profile.

Create variables for at least:

- Name
- Course & Section
- Age
- Hobby
- Height
- Student status

Your data should exist **separately from your UI widgets**.

Screenshot:

- Your variables
- Their data types
- Application still running

```
class _MyHomePageState extends State<MyHomePage> {  
  int _counter = 0;  
  String name = '🔥 Cassandra Gayle R. Oraiz 🔥';  
  int age = 20;  
  String courseAndSection = 'BSIT III';  
  String hobby = 'Drawing, Gaming, Crochet';  
  String favoriteFoods = 'Palabok';  
  String favoriteMovie = '3 Idiots';  
  String favoriteGame = 'Mobile Legends';  
  double height = 149.86;  
  bool isStudent = false;  
  
  void toggleSwitch(){  
    setState(() {  
      isStudent = !isStudent;  
    });  
  }  
}
```

🚩 Flag 3 — Connect Data to UI

Objective: Replace your hardcoded information with your variables.

Your profile should now display information using the variables you created.

Test:

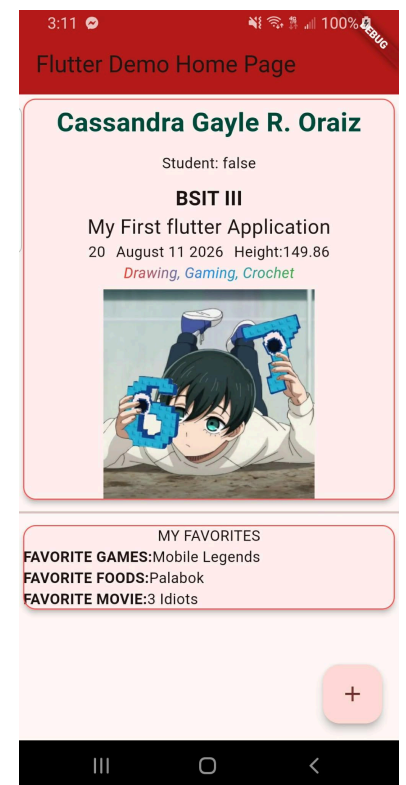
Change the value of one of your variables.

The UI should automatically display the new value.

Screenshot:

- Variables
- Widgets displaying the variables
- Updated application

```
), // Row
RainbowText(
  hobby,
  colors: [Colors.red, Colors.blue, Colors.green],
  style: TextStyle(fontStyle: FontStyle.italic),
), // RainbowText
SizedBox(height: 5),
Image.asset('assets/67.jpg', height: 200, width: 200),
```



🚩 Flag 4 — Make the Image Data-Driven

Objective: Make your profile image part of your data.

Create a variable containing the path to your profile image.

Use the variable when displaying the image.

Then change the variable to another registered image.

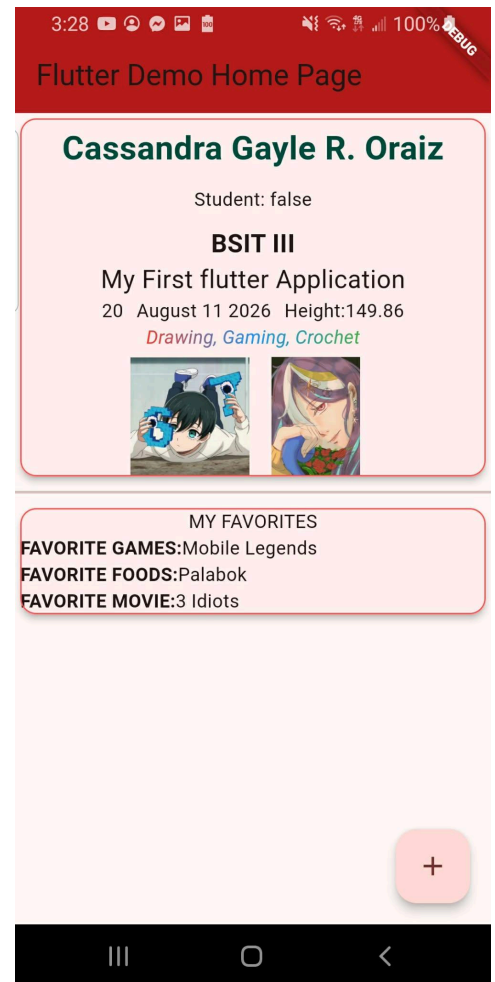
The image displayed by the application should change.

Screenshot:

- Image variable
- Image widget using the variable
- Updated application

```
String profileImage = 'assets/67.jpg';  
String pfp = 'assets/Ran.jpg';
```

```
), // RainbowText  
  SizedBox(height: 5),  
  
  Row(  
    mainAxisAlignment: .center,  
    children: [  
      Image.asset(profileImage, height: 150, width: 150),  
      SizedBox(width: 5),  
      Image.asset(pfp, height: 150, width: 150),  
    ],  
  ), // Row
```





Flag 5 — Five Profiles, One Design

Objective: Create at least **five different profile cards** using the same visual design.

Each profile must contain at least:

- Image
- Name
- Course & Section
- Age
- Hobby

The five profiles must contain different information.

Important

All five profiles should use the **same visual design**.

You are demonstrating that the same UI can represent different data.

Screenshot:

- Your 5 different profile data

```
class _MyHomePageState extends State<MyHomePage> {  
  int _counter = 0;  
  String name = '🔥 Cassandra Gayle R. Oraiz 🔥';  
  int age = 20;  
  String courseAndSection = 'BSIT III';  
  String hobby = 'Drawing, Gaming, Crochet';  
  String favoriteFoods = 'Palabok';  
  String favoriteMovie = '3 Idiots';  
  String favoriteGame = 'Mobile Legends';  
  double height = 149.86;  
  bool isStudent = true;  
  
  String profileImage = 'assets/Me.jpg';  
  
  void toggleSwitch() {  
    setState(() {  
      isStudent = !isStudent;  
    });  
  }  
}
```

```
String name1 = '🍪 Pure Vanilla Cookie 🍪';  
int age1 = 20;  
String courseAndSection1 = 'BSIT III';  
String hobby1 = 'Gardening';  
double height1 = 6.0;  
bool isStudent1 = true;  
  
String profileImage1 = 'assets/PV.jpg';  
  
void toggleSwitch1() {  
  setState(() {  
    isStudent = !isStudent;  
  });  
}  
  
String name2 = '🍷 Rin Itoshi 🍷';  
int age2 = 22;  
String courseAndSection2 = 'BSIT III';  
String hobby2 = 'Football, Horror Enthusiast';  
double height2 = 6.0;  
bool isStudent2 = true;  
  
String profileImage2 = 'assets/67.jpg';  
  
void toggleSwitch2() {  
  setState(() {  
    isStudent = !isStudent;  
  });  
}
```

```
String name3 = '🌟 Theodosia "Ran" Agapov 🌟';  
int age3 = 23;  
String courseAndSection3 = 'BSIT III';  
String hobby3 = 'Football, Hairstyling';  
double height3 = 6.3;  
bool isStudent3 = true;  
  
String pfp3 = 'assets/Ran.jpg';  
  
void toggleSwitch3() {  
  setState(() {  
    isStudent = !isStudent;  
  });  
}  
  
String name4 = '🍷 Shadow Milk Cookie 🍷';  
int age4 = 25;  
String courseAndSection4 = 'BSIT III';  
String hobby4 = 'Chismis';  
double height4 = 6.0;  
bool isStudent4 = false;  
  
String profileImage4 = 'assets/SMC.jpg';  
  
void toggleSwitch4() {  
  setState(() {  
    isStudent = !isStudent;  
  });  
}  
  
void _incrementCounter() {  
  setState(() {  
    _counter++;  
  });  
}
```

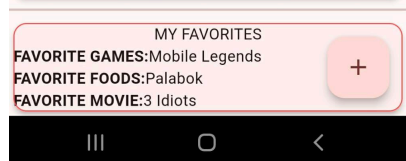
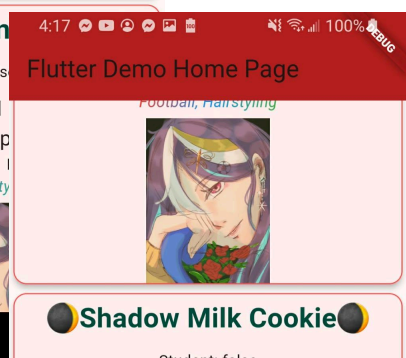
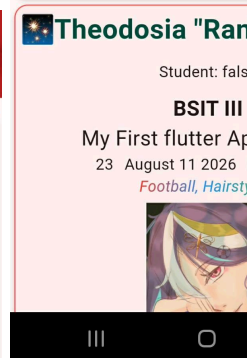
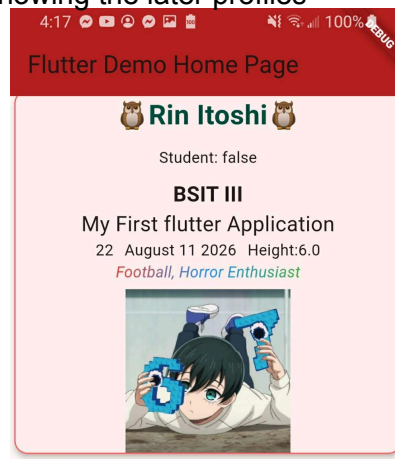
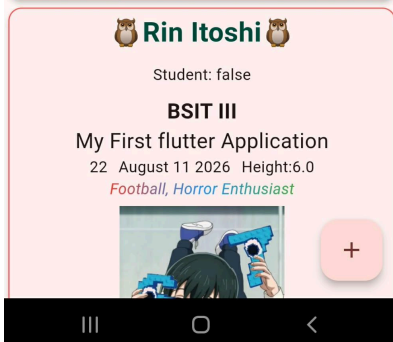
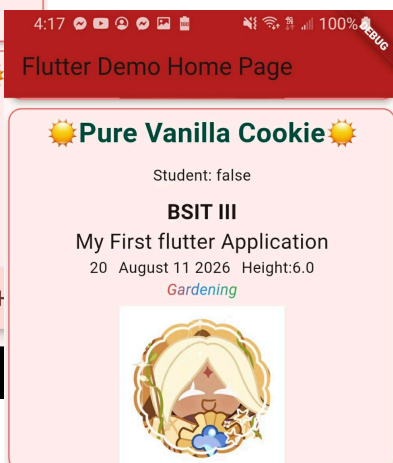
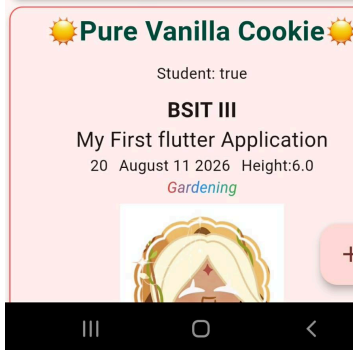
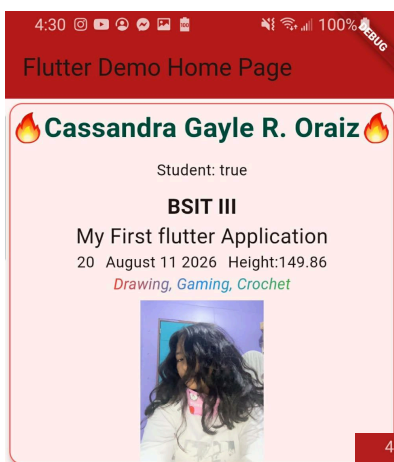
🚩 Flag 6 — THE HIDDEN PROBLEM

You didn't see your UI?

Objective: Make all five profiles accessible to the user.

Screenshot:

- The scrolling widget in your code
- Top portion of the application
- The rest of the portion of the application showing the later profiles



```
body: SingleChildScrollView(  
  child: Column(  
    mainAxisAlignment: .start,  
    children: [  
      Card( // Card ...  
      Card( // Card ...  
      Card( // Card ...  
      Card( // Card ...  
      Card( // Card ...
```

Flag 7 — Survive the Database

Objective: Prepare your UI for incomplete data.

Now imagine your profiles aren't manually created.

Someday, the data will come from a database.

And real data isn't always perfect.

Some information may be:

- Missing
- Empty
- `null`
- Not provided

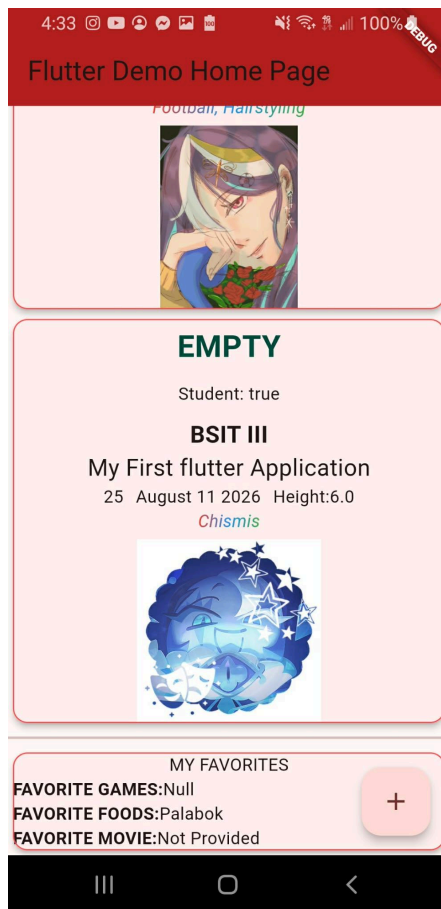
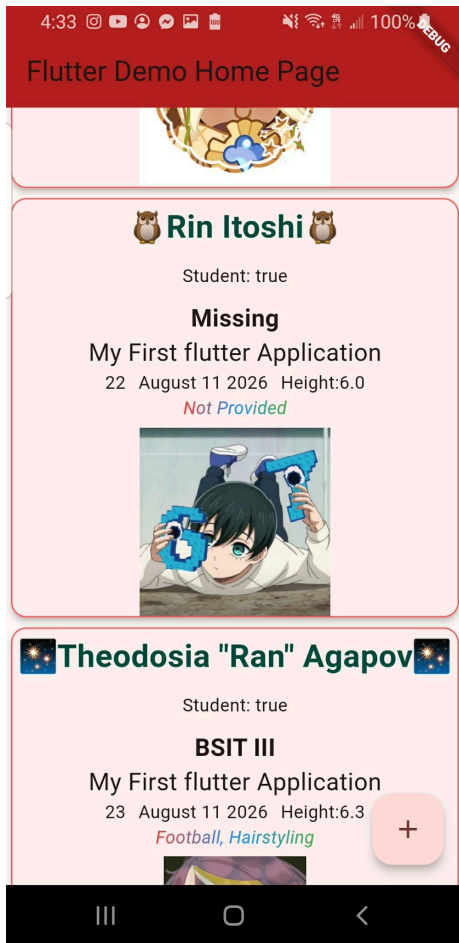
Create at least **three profiles with missing information**.

Challenge

Your placeholders should look intentional and clean.

Screenshot:

- Incomplete profile data
- Application displaying fallback values
- Application still functioning correctly





Flag 8 — Save Your Progress

Objective: Commit and push your completed Data-Driven UI CTF.

Use an appropriate commit message following proper Git standards.

Branch:

feature/data-ui

Screenshot:

- Successful commit
- Current branch
- Terminal showing the commit

```
oraiz@Mooney MINGW64 ~/Mynewflutterapp/IT314AB_ORAIZ (feature/data-ui)
$ git status
On branch feature/data-ui
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
    modified:   my_first_flutter_app/lib/main.dart
    modified:   my_first_flutter_app/pubspec.yaml

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    my_first_flutter_app/assets/Me.jpg
    my_first_flutter_app/assets/PV.jpg
    my_first_flutter_app/assets/Ran.jpg
    my_first_flutter_app/assets/SMC.jpg

no changes added to commit (use "git add" and/or "git commit -a")

oraiz@Mooney MINGW64 ~/Mynewflutterapp/IT314AB_ORAIZ (feature/data-ui)
$ git add .

oraiz@Mooney MINGW64 ~/Mynewflutterapp/IT314AB_ORAIZ (feature/data-ui)
$ git commit -m "Feat: Added more profiles, Made app able to scroll, Simulated missing data, Added more data"
[feature/data-ui 5d615ce] Feat: Added more profiles, Made app able to scroll, Simulated missing data, Added more data
6 files changed, 336 insertions(+), 50 deletions(-)
create mode 100644 my_first_flutter_app/assets/Me.jpg
create mode 100644 my_first_flutter_app/assets/PV.jpg
create mode 100644 my_first_flutter_app/assets/Ran.jpg
create mode 100644 my_first_flutter_app/assets/SMC.jpg
```